

Course code: ME332

Course Name: Thermal Management of Electronic System

Pre-requisite: Engineering Thermodynamics (ME 103)

Objective of the course:

To provide detailed knowledge about IC Engines and Power Plant

Course Content:

UNIT I: Thermodynamic Cycles for Energy Conversion (3 Lectures)

Gas power cycles, vapour power cycles, refrigeration cycles

UNIT I : Introduction & Fuels for SI and CI engine (5 Lectures)

Basic Nomenclature, Engine classification, working principle of 2 stroke and 4 stroke, fuel air cycles and actual cycles, valve and port timing diagram. Classification of fuels, fuel properties and fuel rating, mixture requirements

UNIT II : Combustion in SI and CI Engine (5 Lectures)

Stages of combustion in SI and CI engine, detonation, effect of engine design and operating variables on engine detonation, combustion knock, effect of engine design and operating variables on engine knock, comparison of knock in SI and CI engine.

UNIT III : Carburettor and Fuel Injection system & Engine friction and lubrication (5 Lectures)

Types of carburettor, calculation of A/F ratio for simple carburettor, mixture requirements, injection system in SI and CI engine, components of engine friction, friction mean effective pressure. Lubrication principle, types of lubrication, properties of lubricant.

UNIT IV : Heat transfer and Cooling system & Engine Testing and performance (6 Lectures)

Engine temperature distribution, gas temperature variation, air cooling and liquid cooling systems. Measurement of indicated power, brake power, fuel consumption, air flow rate, speed, heat balance sheet..

UNIT V : Introduction & Power Plant Thermodynamics (5 Lectures)

History and present status of energy resources. Energy sources for modern plants. Power systems and utility demand patterns. Economics of electric power production, Basic laws of thermodynamics. Thermodynamic

power cycles. Steam cycles for modern power plants.

UNIT VI : Fossil-Fueled Steam Power Plants & Combustion (8 Lectures)

Solid, Liquid and gaseous fuel. Properties of fuels. Coal and coal cleaning and processing Introduction to power plant systems and components. Fuel handling and preparation. Steam generator configuration and analysis. Steam separation and purification. Natural convection systems. Steam generator control. Air circulating and heating system. Water treatment systems. Cooling towers. Emission control systems. Waste disposal, Principles of combustion. Combustion calculations. Theoretical and actual flame temperatures. Combustion systems

UNIT VIII: Renewable Energy Sources (2 Lectures)

Basic concepts of renewable resources: solar, wind and biomass

Laboratory Component:

1. Performance testing of single cylinder petrol engine
2. Performance testing of multi cylinder petrol engine
3. Performance testing of multi cylinder diesel engine
4. Performance testing of variable compression SI engine
5. Performance testing of variable compression CI engine
6. Emission testing of internal combustion engine
7. Study of cross sectional models of automobile parts
8. Performance characteristics of solar collector
9. Study on renewable energy based electricity generation system

Readings:

Text Books:

1. Fundamentals of Internal Combustion Engine: H.N. Gupta
2. Power Plant Engg: P.K. Nag

Reference Books:

1. Internal Combustion Engine: J.B. Heywood
2. Internal Combustion Engine: V. Ganeshann
3. Fundamentals in power plant Engg., R. Yadav
4. Steam and Its Use: Babcock